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A fast recovery probe and receiver for pulsed nuclear magnetic resonance spectroscopy

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Abstract A single coil probe and receiver for use in pulsed nuclear magnetic resonance experiments are described. The impedance transformation properties of a one-quarter wavelength line are used to protect the receiver during the radio-frequency pulse. The probe is very efficient in its use of r.f. power to create an oscillating magnetic field, and the overall system has a fast recovery time (less than $4 \mu\text{s}$ at 30 MHz).

1 Introduction

Various techniques for designing sample probes for use in pulsed n.m.r. measurements on solids are described in the literature. A review of these methods may be found in papers by Clark (1964), Lowe and Baraal (1963), Mansfield and Powles (1963), McKay and Woessner (1966) and Trappaniers *et al.* (1964) and in the references cited in these works. Each of these methods, however, is deficient in one or more of the following properties: recovery time, efficient impedance matching to the appropriate r.f. transmitter and/or receiving amplifier, ease of use and suitability for use over a wide temperature range.

This paper describes a simple single coil probe and receiver system that is useful over a wide temperature range. This system provides short recovery times ($4 \mu\text{s}$ at 30 MHz), ease of tuning and good impedance matching.

The parameters in the design described here are adjusted for operation at a frequency of 30 MHz, but they may be easily changed for operation at other frequencies.

2 Circuit description

Figure 1 shows a block diagram of the probe and associated components.

The output impedance of the transmitter is matched to the transmission line and is equal to the input impedance of the receiver. The impedance at resonance of the sample tank circuit is also equal to that of the transmission line.

The operation of the system depends upon the nonlinear properties of semiconductor diodes. Semiconductor diodes

have a sharp corner in their current-voltage characteristic curve. Thus, when a small voltage ($\ll 0.5 \text{ v}$) is applied to a diode, little current flows and the diode appears as a high impedance. When a large voltage ($\gg 0.5 \text{ v}$) is applied, a large current flows, and the diode appears as a low impedance.

During the application of a high-powered r.f. pulse from the transmitter, both sets of crossed diodes in figure 1 are conducting heavily and have low impedances. Thus, the r.f. pulse passes through the diodes at A without appreciable attenuation, while the diodes at B prevent the r.f. level at the receiver input from rising above about 1 v. Since point A is a quarter wavelength from B, the effective short circuit at B appears as an open circuit at A and thus does not affect the voltage at A.

Following the r.f. pulse and after the transients have died down, the diodes stop conducting, and the series crossed diodes present a large impedance at point A, while the impedance at point B becomes the characteristic line impedance. The small signal voltages (here typically about 1 mv) are not large enough to lower the impedance of the diodes. Thus, in the absence of an r.f. pulse, the transmitter is isolated from the system, while the receiver is matched to the sample tank circuit.

It should be observed that the length of the $\lambda/4$ line between points A and B is not especially critical, as the purpose of this line is to allow the use of the diode short at B without presenting a low impedance shunt at the sample coil. Thus,

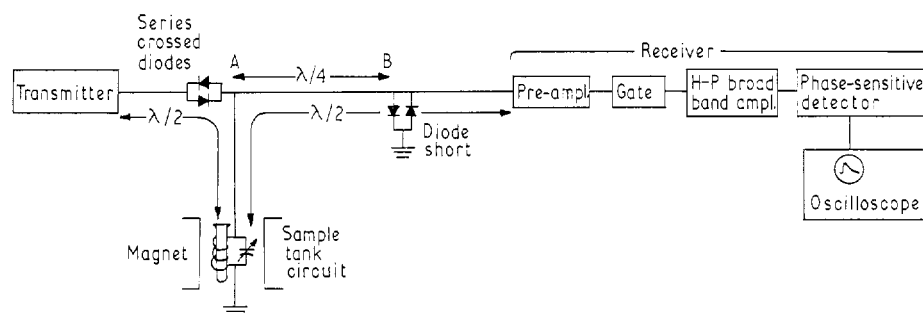


Figure 1 Block diagram of probe and receiver

since the transformed impedance of the diode short need only be large compared with $185\ \Omega$, within the bandwidth of the tuned sample coil, the cable need only be of the order of $\lambda/4$, and the use of the quarter wave cable does not entail any significant narrow-banding of the system.

The electrical lengths of the transmission lines from the transmitter to the sample coil and from the sample coil to the receiver are each one half wavelength. This minimizes reflections due to slight mis-matches in the connections, etc.

Multiple pairs of crossed diodes were used at A and B to reduce their impedances in the conducting state. The series crossed diodes contained 5 pairs of 1N3604's, and the diode short contained 8 pairs.

3 Probe design

Suppose the intrinsic fall time of the transmitter pulse of peak amplitude V_0 into a purely resistive load is shorter than the ringing time constant of the sample tank circuit. Then, if the recovery time of the receiver is shorter than the time required for the r.f. pulse to decay to a value of the order of the induced nuclear signal voltage V_n , the recovery time T_R of the system is given by:

$$T_R = (Q/\pi f) \ln (V_0/V_n) \quad (1)$$

where $Q = R/\omega L$.

For the recovery figures quoted in this work we have $f = 30\text{ MHz}$, $R = Z_0 = 185\ \Omega$, $V_0/V_n \sim 10^7$ and $T_R \sim 3\ \mu\text{s}$, which yields $Q \approx 17$.

The probe consisted of a variable capacitor and a sample coil that were suspended from a Dewar top plate by a rigid coaxial line. The complete mechanical details of the probe construction will not be given here, as the dimensions must be tailored to individual experimental situations.

The capacitor consisted of several copper shim stock plates separated by annealed glass microscope slide covers 0.006 in thick. The plates were rigidly clamped at one end and their separation varied by means of a screw adjustment at the other end. This screw was operated from the top plate by a stainless-steel tube passed through an O-ring seal so that the probe could be operated in a vacuum if necessary.

In order to provide good homogeneity of the oscillating r.f. magnetic field and a good filling factor, the sample coil was close wound over the desired sample dimensions. In order to keep the impedance and Q at the desired values, however, the inductance was kept much lower than is practical for a wire wound coil. For this reason the coil was wound of copper shim stock. In the present work the coil was three turns, wound in a single layer 1 cm in diameter and 1.2 cm in length from 0.007 in copper shim stock, as shown in figure 2. The Q of the coil was adjusted by means of a short length of manganin wire placed in series with the winding so that Q was approximately constant from 1°K to 300°K . A $\frac{1}{4}$ in length of no. 20 (0.032 in) manganin wire was found to be satisfactory in producing the required Q of about 17.

To ensure mechanical rigidity, the coil was potted in sulphur. This was accomplished by winding the coil on a Teflon mandrel. The mandrel was placed in a Teflon sleeve with an inner diameter about twice that of the coil, and the resulting annulus was filled with molten sulphur. Sulphur was chosen as a potting agent as it has no nuclei with a magnetic moment.

The coil leads were flat copper strips that were brought out of the coil parallel and close together to reduce lead inductance. The insulation necessary to separate the leads was glass filter paper, as shown in figure 2.

Finally, the coil was enclosed in a brass shield about 2.5 cm

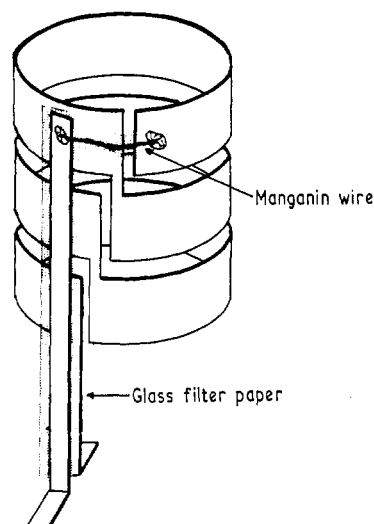


Figure 2 Coil construction

in diameter and about 7 cm in length for electrical shielding and mechanical rigidity.

4 Receiver

4.1 Preamplifier

A two stage preamplifier with a recovery time of about $1\ \mu\text{s}$ from an overload of 1 v and with an input impedance of $185\ \Omega$ was constructed for use in the system. The preamplifier was stagger tuned to provide a flat response over a bandwidth of 4 MHz (3 db down), centred at 30 MHz. The resulting voltage gain was 135, and the noise figure less than 3.5 db.

A standard 30 MHz preamplifier (Linear Electronics Laboratories, with a voltage gain of 1000) was also tested in the system. However, it was not entirely satisfactory, because of small amplitude ringing following partial recovery from an r.f. pulse. This ringing was due to decoupling chokes used in the B^+ lines between stages.

4.2 Diode gate

The broadband amplifier (Hewlett-Packard 461-A) recovers in less than $1\ \mu\text{s}$ from the overload presented by the r.f. pulse feedthrough, but the gain is rather unstable for about $10\ \mu\text{s}$ following the pulse. To prevent distortion of the free induction decay resulting from this gain instability, a balanced diode gate was used to decouple the preamplifier from the broadband amplifier for the duration of the r.f. pulse and for a short time thereafter. The gate was placed between the preamplifier and the amplifier so that the noise generated in the diodes by the current passed through them in their conducting state would be injected at a relatively high signal level.

Since the diode gate is completely balanced, it does not produce any ringing or blocking effects due to turn-on transients. Also, since the gate prevents overloads from the r.f. pulses from reaching the detector, the system is especially well suited to use with signal averaging equipment, as there are no large transients to interfere with the analogue-to-digital converter.

The gate circuit is shown in figure 3. When the gate is in its normal (closed) stage, side C of the diode bridge is negative with respect to side D. Thus, current flows around the diode bridge, and a signal can pass from the input branch to the output branch. When a positive pulse (+4 v minimum) is applied to the phase splitting network, the transistors conduct, and the bias on the bridge is reversed, and no signal can pass the gate.

The 1 k Ω potentiometer across the gate provides d.c.

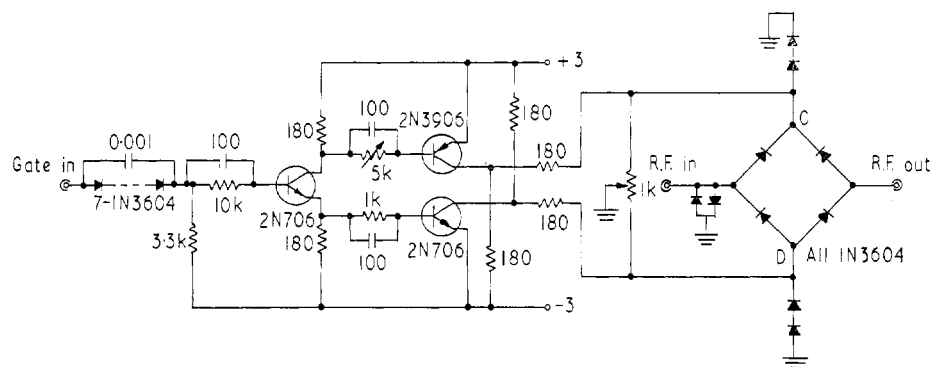


Figure 3 Schematic diagram of diode gate used to protect broadband amplifier

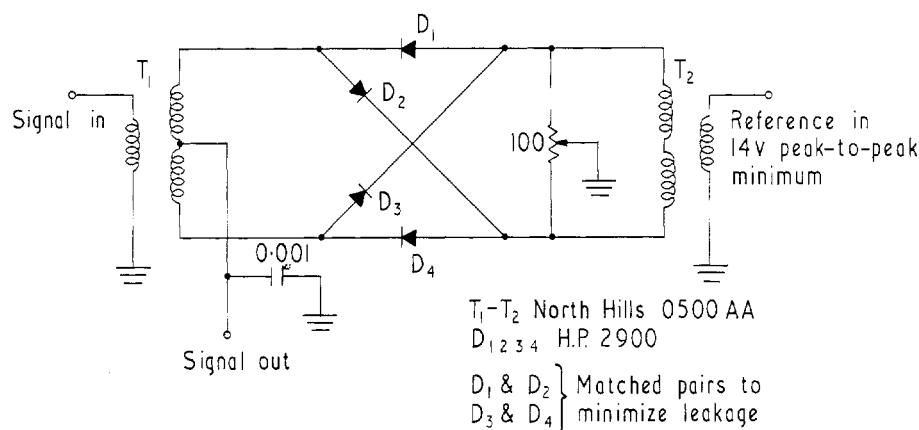


Figure 4 Schematic diagram of phase sensitive detector

balance for the gate. The 5 k Ω potentiometer in the base of the 2N3906 sets the drive to that transistor and is set to balance out the turn-on transient of the bridge. The feedthrough of the r.f. pulse can be seen at the beginning of figure 5 and is less than one tenth the amplitude of the detected signal. This feedthrough of the r.f. pulse to the input of the broadband amplifier is less than 0.010 v peak-to-peak over a broad range (500–1200 v peak-to-peak) of r.f. pulse amplitudes. The turn-on transients of the gate may be balanced to less than ± 0.001 v.

4.3 Phase sensitive detector

The circuit of the phase sensitive detector is shown in figure 4. The circuit has a flat frequency response (0.1 dB) from 1 to 100 MHz. This broadband response is provided by the two North Hills 0500 AA transformers (North Hills Electronics Inc., Glen Cove, L.I., New York), which have a flat frequency response (0.1 dB) from under 1 MHz to over 100 MHz (down 1 dB at approx. 0.1 MHz and 200 MHz) and have an input impedance of 50 Ω (unbalanced) and an output impedance of 200 Ω .

The detector requires a minimum r.f. reference level of 14 v peak-to-peak and has a linear response for input signal levels to 1.2 v. With no filtering, the r.f. leakage can be set to less than 0.01 v peak-to-peak and the d.c. offset to less than ± 0.01 v with the 100 Ω balancing potentiometer.

The input and output impedances of the phase sensitive detector are about 50 Ω .

5 Conclusion

The probe and receiver system described above are free from many of the difficulties inherent in most other systems, as there are no balancing or damping elements. Also, since the

entire system is impedance matched to the single coil probe without the use of transformers, there are no difficulties with transformer ringing, and there is optimum r.f. power transfer. The quarter wavelength cable and diode short effectively prevent the r.f. transmitter pulse from blocking the receiver.

Acknowledgments

We wish to thank D Douglas of the Bell Telephone Laboratories for suggesting the use of a quarter wavelength cable to isolate the receiver from the transmitter, and W Vollmers for building the r.f. probe.

It has been brought to our attention that a somewhat similar use of a quarter wavelength cable is described by U Häberlen (1967 Ph.D. Thesis, Tech. Hochschule, Stuttgart).

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References

- Clark W G 1964 *Rev. Sci. Instrum.* **35** 316–33
- Lowe I J and Barnaal D E 1963 *Rev. Sci. Instrum.* **34** 143–6
- Mansfield P and Powles J G 1963 *J. Sci. Instrum.* **40** 232–8
- McKay R A and Woessner D E 1966 *J. Sci. Instrum.* **43** 838–40
- Trappaniers N J Gerritsma C J and Osting P H 1964 *Physica* **30** 997–1017